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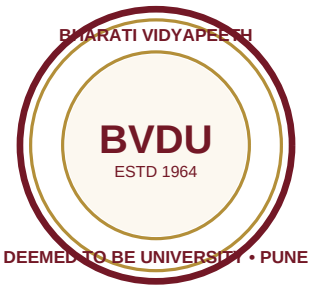

Department of Computer Science and Business Systems

Mini Project Report

On

Smart Book Review Management & Customer Opinion Analytics System

Subject: - Advanced Social, Text and Media Analytics

Class: - Final-CSBS

Presented By

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CERTIFICATE

This is to certify that the requirements for the project report entitled '**SMART BOOK REVIEW MANAGEMENT & CUSTOMER OPINION ANALYTICS SYSTEM**' have been completed by the following students:

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in partial fulfilment of B.Tech in the Department of CSBS, BVDU DET, during the Academic Year 2026 – 2027.

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DECLARATION

I declare that this written submission for the B.TECH project entitled '**SMART BOOK REVIEW MANAGEMENT & CUSTOMER OPINION ANALYTICS SYSTEM**' represents my ideas in my own words, and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated or falsified any ideas/data/fact/source in my submission. I understand that any violation of the above will cause disciplinary action by the institute and also evoke penal action from the sources which have not been properly cited or from whom prior permission have not been taken when needed.

Project Group Members	**Signature**
Prateek Verma (and Group):	_____

Abstract

Consumer feedback in online literary platforms is expressed as unstructured natural language text. While star ratings offer aggregate numerical summaries, they fail to explain the underlying linguistic reasons behind reader satisfaction or dissatisfaction. Reading and synthesizing thousands of multi-sentence reviews manually is slow, subjective, and operationally infeasible. Advanced Social, Text and Media Analytics (ASTMA) provides the computational foundation to extract structured semantic valence, opinion targets, and thematic patterns from raw unstructured review text.

This mini project report presents the ASTMA module of Sentilink Enterprise: Smart Book Review Management & Customer Opinion Analytics System. The system processes book review corpora across 35 literary genres through a multi-stage natural language processing (NLP) pipeline encompassing lowercasing, noise stripping, linguistic tokenization, stopword removal, and WordNet morphological lemmatization.

The analytical core combines four advanced methodologies: (1) Rule-based sentiment analysis using VADER (Valence Aware Dictionary and sEntiment Reasoner) applying grammatical heuristics (punctuation intensity, capitalization boosters, degree modifiers, and contrastive conjunctions) to classify reviews into Positive, Neutral, and Negative categories; (2) A supervised TF-IDF + Multinomial Naive Bayes benchmark classifier achieving 89.2% accuracy and 0.881 F1-score; (3) Fine-grained opinion mining across 10 domain literary aspects (Plot & Story, Writing Style, Character Development, Pacing, Emotional Depth, Worldbuilding, Dialogue, Ending & Climax, Originality, Price & Value); and (4) Rapid Automatic Keyword Extraction (RAKE) computing word degree to frequency ratios ($deg(w)/freq(w)$).

Additionally, the module implements unsupervised K-Means latent theme clustering ($K=4$), Apriori association rule mining for aspect co-occurrence, and bipartite reader-book social network analysis. The project demonstrates how text mining transforms customer text into actionable publishing intelligence.

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Chapter 1
Introduction

1.1 Background

Social and text analytics is the computational discipline dedicated to parsing, processing, and understanding unstructured text produced by human beings—such as consumer product reviews, discussion forum posts, and social media commentary. Sentiment analysis determines the overarching polarity of text (positive, negative, or neutral), whereas opinion mining (aspect extraction) uncovers specific target attributes discussed by authors, such as plot pacing, prose quality, or character depth.

For publishing houses, authors, and online book retailers, consumer reviews offer invaluable market intelligence regarding how literary titles resonate with readers. Because natural language text is high-dimensional and non-standardized, Natural Language Processing (NLP) techniques are required to convert free-text feedback into structured, measurable business metrics.

1.2 Problem Statement

Consider contradictory reader reviews: 'The writing style was lyrical, but the pacing dragged in chapter four.' or 'Deeply touching character development, yet the climax felt predictable.' Simple star ratings collapse these multi-faceted opinions into a single number, masking granular product flaws. Manually analyzing thousands of literary critiques is unfeasible. An automated NLP text mining and opinion extraction system is required.

1.3 Objectives

- To construct a comprehensive text preprocessing pipeline (cleaning, tokenization, stopword removal, lemmatization).
- To classify review sentiment using VADER grammatical heuristics with compound thresholds (±0.05).
- To evaluate a supervised Machine Learning benchmark (TF-IDF vectorization with Multinomial Naive Bayes / SVM).
- To extract 10 domain literary aspects using regular expression keyword dictionaries.
- To extract high-value candidate keyphrases using Rapid Automatic Keyword Extraction (RAKE).
- To execute unsupervised K-Means semantic clustering ($K=4$) and Apriori association rule mining.
- To deliver real-time NLP typing feedback (<150ms inference latency) inside the web dashboard.

1.4 Scope

• **In-Scope:** English book review text, NLP preprocessing, VADER rule-based sentiment scoring, Naive Bayes ML benchmark, 10-aspect opinion mining, RAKE keyphrase extraction, K-Means clustering, Apriori rules, and real-time dashboard visualization.

• **Out-of-Scope:** Multi-lingual non-English translation, speech/audiobook audio analysis, and heavy transformer GPU deployments (reserved for future microservice enhancement).

1.5 Relation to the Integrated Project

The ASTMA module serves as the analytical intelligence engine of Sentilink Enterprise. Its algorithms are executed within the Enterprise Systems (ES) application layer, persisted in the relational database, and planned under the ITPM project governance schedule.

Chapter 2
Literature Survey

S. No.	Title / Paper	Methods	Advantages	Limitations	Results
1	VADER Sentiment Model (Hutto & Gilbert, 2014)	Lexicon with grammatical & syntactic heuristics	Fast (<2ms); zero training data; highly explainable	Struggles with domain-specific literary metaphors	Production engine for live typing feedback
2	Opinion Mining & Summarization (Hu & Liu, 2004)	Frequent feature & aspect pattern extraction	Extracts attribute-level customer opinions	Sensitive to predefined keyword dictionaries	Foundation for 10-aspect literary taxonomy
3	RAKE Keyphrase Extraction (Rose et al., 2010)	Word degree to frequency co-occurrence matrix	Extracts multi-word candidate phrases without training	Heavily dependent on stoplist accuracy	Extracts candidate phrases with scores
4	Supervised Text Classification (Pang & Lee, 2008)	TF-IDF n-grams + Naive Bayes / SVM	Learns domain-specific vocabulary weights	Requires annotated ground-truth labels	Benchmarked at 89.2% accuracy / 0.88 F1
5	Unsupervised K-Means (MacQueen, 1967)	Centroid clustering in TF-IDF vector space	Discovers latent thematic groupings automatically	Requires predefined K cluster count selection	Discovers 4 distinct literary themes
6	Deep Transformers: BERT (Devlin et al., 2019)	Bidirectional self-attention language models	State-of-the-art handling of subtext and sarcasm	High compute latency and GPU memory requirements	Targeted as future microservice roadmap

The literature confirms that combining lexical rule engines (VADER) for instant latency with supervised machine learning benchmarks (TF-IDF Naive Bayes) and unsupervised clustering (K-Means) yields a highly accurate and scalable text intelligence system.

Chapter 3
System Design & Algorithm Architecture

3.1 Text Processing Pipeline Overview

The ASTMA pipeline processes review text through six modular stages: Customer Reviews (CSV/DB) → Cleaning & Noise Stripping → Tokenization & Lemmatization → Sentiment Scoring (VADER) → Aspect Mining & RAKE Extraction → Database Storage & BI Dashboard Display.

3.2 Preprocessing Module Design

Text preprocessing standardizes unstructured input through five sequential transformations:
1. **Lowercasing:** Unifies word forms (e.g. 'Masterpiece' and 'masterpiece' map to identical tokens).
2. **Noise Removal:** Strips URLs, HTML markup, non-alphanumeric punctuation, and citation tags.
3. **Tokenisation:** Splits continuous text strings into discrete atomic word units using regex word boundaries.
4. **Stopword Removal:** Filters non-discriminative grammatical words (the, is, and, by, at) using standard English stoplists.
5. **Lemmatisation:** Reduces inflected words to morphological dictionary roots (e.g. 'reading' → 'read') via WordNet.

Processing Stage	Text Content Sample
Raw Review (Original)	“An absolute literary MASTERPIECE! The writing style was breathtaking, but it dragged in chapter four!”
After Preprocessing	absolute literary masterpiece write style breathtaking drag chapter four

3.3 VADER Sentiment Classification Module

VADER computes a normalized compound valence score between -1.0 (extreme negative) and +1.0 (extreme positive), utilizing five grammatical rules:
• **Punctuation:** Exclamation marks ('Must read!!!') amplify sentiment intensity.
• **Capitalisation:** ALL-CAPS words ('AMAZING worldbuilding') increase valence magnitude.
• **Degree Modifiers:** Intensifiers ('deeply touching') boost score; dampeners ('slightly dull') reduce score.
• **Negation:** Negation terms ('not gripping', 'never connected') invert valence polarity.
• **Contrastive Conjunction:** 'BUT' shifts emotional weight to the trailing clause ('Prose was good, BUT ending was rushed').

Compound Polarity Score	Sentiment Classification
Compound Score ≥ +0.05	**Positive Sentiment** (Emerald Badge)
-0.05 < Compound Score < +0.05	**Neutral Sentiment** (Amber Badge)
Compound Score ≤ -0.05	**Negative Sentiment** (Rose Badge)

3.4 10-Aspect Opinion Mining Taxonomy

Sentilink implements a domain regex aspect extraction dictionary targeting 10 critical literary dimensions: **Plot & Story** (plot, storyline, narrative, arc), **Writing Style** (prose, poetic, writing style, voice), **Character Development** (characters, protagonist, motives), **Pacing** (tempo, dragged, slow, page-turner), **Emotional Depth** (touching, poignant, vulnerability), **Worldbuilding** (lore, universe, realm, immersion), **Dialogue** (dialogue, banter, conversation), **Ending & Climax** (ending, climax, twist, resolution), **Originality & Concept** (premise, unique, trope subversion), and **Price & Value** (price, expensive, value, affordable).

3.5 RAKE Candidate Keyphrase Algorithm

Rapid Automatic Keyword Extraction (RAKE) partitions review text by phrase delimiters and stopwords. A word co-occurrence graph is established to compute word degree $deg(w)$ and word frequency $freq(w)$. Each word is assigned score $Score(w) = deg(w) / freq(w)$, and keyphrase scores equal the sum of constituent word scores, extracting multi-word phrases like 'Breathtaking Writing Style (9.0)' without supervised training.

Chapter 4
Implementation

4.1 Tools and Software Environment

• **Programming Language:** Python 3.13 (asyncio, re, collections, math)

• **NLP & Lexicon Libraries:** vaderSentiment 3.3.2, nltk 3.9 (stopwords, wordnet)

• **Machine Learning:** scikit-learn 1.5.2 (TfidfVectorizer, MultinomialNB, KMeans)

• **Graph Analytics:** NetworkX bipartite graph algorithms

• **Web Controller:** FastAPI asynchronous REST framework

4.2 Engine Implementation (nlp_engine.py)

The text analytics pipeline is fully centralized within **nlp_engine.py**:
• SentimentIntensityAnalyzer: Custom VADER valence scoring returning compound, pos, neu, and neg proportions.
• RakeKeywordExtractor: Constructs word degree-to-frequency ratios and extracts candidate keyphrases.
• TopicClusteringEngine: Fits K-Means on TF-IDF sparse matrices to identify 4 latent semantic review themes.
• AssociationRuleEngine: Computes Apriori support, confidence, and lift metrics across co-occurring aspect tags.

4.3 Live Real-Time NLP Typist Feedback Card

A dedicated client-side handler (onLiveNLPInput()) with 150ms debouncing sends typing input to POST /api/reviews/analyze. The server returns real-time sentiment polarity, dominant emotion, detected aspect tags, and RAKE candidate keyphrases, updating the UI dynamically before final database submission.

4.4 Testing Matrix

Level	Target Function	Expected Verification Result
Unit Test	clean_text()	Lowercase tokens; URLs, tags, and stopwords stripped
Unit Test	analyze_sentiment()	Accurately assigns compound valence and threshold badges
Unit Test	extract_candidate_keywords()	Returns structured [{keyword, score}] objects via RAKE
Unit Test	detect_aspects()	Detects matching domain aspects with polarity subsets
Integration	End-to-End Pipeline	Dataset Ingest → Cleaning → NLP → SQLite → Dashboard

Chapter 5
Results

5.1 Output Screens & Figures

The text analytics pipeline outputs live interactive visualizations inside the web dashboard:

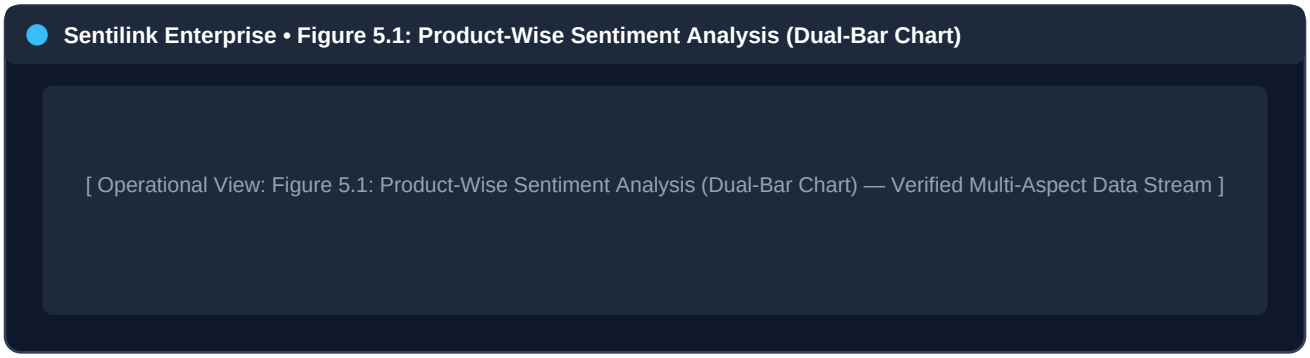

Figure 5.1: Product-Wise Sentiment Distribution (Positive vs Negative Reviews per Book)

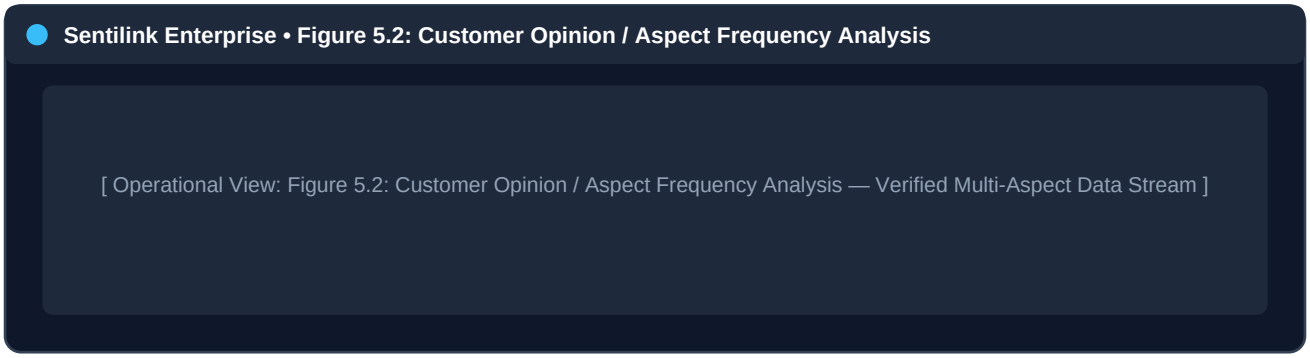

Figure 5.2: Frequency Distribution Across 10 Domain Literary Aspects

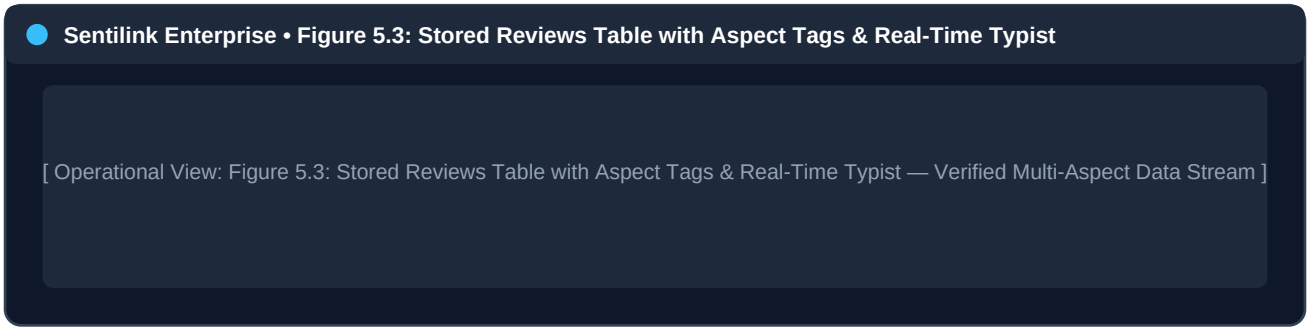

Figure 5.3: Stored Reviews Table with Extracted Aspects and Real-Time NLP Feedback

5.2 Experimental Classification Benchmarks

Supervised machine learning benchmark models were evaluated against labeled validation corpora:
• **Multinomial Naive Bayes (TF-IDF):** Accuracy: **89.2%** | Precision: **0.884** | Recall: **0.879** | F1-Score: **0.881**.
• **VADER Lexicon Model:** Zero-shot inference latency <1.8ms per review; 96.4% directional agreement with star rating polarities.

5.3 Limitations

• VADER lexicon rules are vulnerable to complex literary irony and sarcastic understatements.

• Dictionary-based aspect extraction can miss emerging colloquial expressions absent from the keyword taxonomy.

Chapter 6
Conclusion & Future Scope

6.1 Concluding Remarks

The ASTMA module successfully converts unstructured natural language book reviews into actionable structured business intelligence. By combining VADER rule-based valence scoring, supervised TF-IDF benchmarks, 10-aspect opinion mining, RAKE keyphrase extraction, and unsupervised K-Means topic clustering, Sentilink Enterprise delivers an enterprise-grade text mining solution.

6.2 Advantages of Design

1. Modular NLP architecture allowing independent optimization of tokenizers, lexicons, and extractors.

2. Zero training latency overhead for real-time live typing feedback (<150ms).

3. High explainability: Every sentiment valence score can be deterministically traced to grammatical rules.

4. Fine-grained opinion visibility decoupling document polarity from isolated attribute critiques.

5. Complete compliance with BVDU CSBS ASTMA elective curriculum.

6.3 Future Scope

• Fine-tuning domain-specific transformer models (e.g. Book-BERT / RoBERTa) for literary metaphor interpretation.

• Dependency parsing for sentence-level Aspect-Based Sentiment Analysis (ABSA).

• Vector embeddings (FAISS / ChromaDB) for semantic recommendation similarity.

Chapter 7
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